

CSEC Mathematics
June 2016 – Paper 2
Solutions

SECTION I

Answer ALL questions in this section.

All working must be clearly shown.

1. (a) Using a calculator, or otherwise, calculate

(i) $\frac{3\frac{3}{8} - 2\frac{1}{4}}{1\frac{1}{2}}$ giving your answer as a fraction in its lowest terms. [2]

$$\text{Numerator} = 3\frac{3}{8} - 2\frac{1}{4}$$

$$= \frac{27}{8} - \frac{9}{4}$$

$$= \frac{27-18}{8}$$

$$= \frac{9}{8}$$

$$\text{Denominator} = 1\frac{1}{2}$$

$$= \frac{3}{2}$$

Now,

$$\text{Numerator} \div \text{Denominator} = \frac{9}{8} \div \frac{3}{2}$$

$$= \frac{9}{8} \times \frac{2}{3}$$

$$= \frac{3}{4}$$

$$\therefore \frac{3\frac{3}{8} - 2\frac{1}{4}}{1\frac{1}{2}} = \frac{3}{4}$$

- (ii) $(2.86 + 0.75) + 0.481^2$ giving your answer correct to 2 decimal places.
[2]

$$\begin{aligned}(2.86 + 0.75) + 0.481^2 &= 3.61 + 0.231361 \\ &= 3.841361 \\ &= 3.84 \quad (\text{to 2 decimal places})\end{aligned}$$

(b) Paul bought and sold a computer. He wrote his business activity as follows:

Cost price of computer	=	\$1064
Marked price of computer	=	\$1399
Discount on marked price	=	5%
(if paid by cash)		

Calculate

- (i) the selling price (paid cash) [2]

$$\begin{aligned}\text{Discount} &= 5\% \text{ of cash payment} \\ &= 0.05 \times 1399 \\ &= \$69.95\end{aligned}$$

$$\begin{aligned}\text{Selling Price} &= \text{Marked price} - \text{Discount} \\ &= \$1399 - \$69.95 \\ &= \$1329.05\end{aligned}$$

$\therefore$ The selling price is \$1329.05.

- (ii) the profit or loss as a percentage of the cost price.

$$\begin{aligned}\text{Profit} &= \$1329.05 - \$1064.00 \\ &= \$265.05\end{aligned}$$

$$\begin{aligned}\text{Percentage profit} &= \frac{\text{Profit}}{\text{Cost price}} \times 100 \\ &= \frac{\$265.05}{\$1064} \times 100 \\ &= 24.9\% \quad (\text{to 3 significant figures})\end{aligned}$$

$\therefore$ The profit as a percentage of the cost price is 24.9%.

If cash was not paid, then a discount is not applied.

$$\begin{aligned}\therefore \text{Profit} &= \$1399 - \$1064 \\ &= \$335\end{aligned}$$

$$\begin{aligned}\text{Profit percentage} &= \frac{\text{Profit}}{\text{Cost price}} \times 100 \\ &= \frac{\$335}{\$1064} \times 100 \\ &= 31.5\%\end{aligned}$$

(c) Orange juice is sold in cartons of three different sizes.

Carton Size	Selling Price
350 ml	\$4.20
450 ml	\$5.35
500 ml	\$5.80

Which size of orange juice is the most cost-effective buy?

Justify your answer.

[3]

Carton Size of 350 ml = \$4.20

$$1 \text{ ml} = \frac{\$4.20}{350}$$

$$= \$0.012$$

Carton Size of 450 ml = \$5.35

$$1 \text{ ml} = \frac{\$5.35}{450}$$

$$= \$0.0119$$

Carton Size of 500 ml = \$5.80

$$1 \text{ ml} = \frac{\$5.80}{500}$$

$$= \$0.0116$$

The most cost-effective carton would be the 500 ml, as it costs least per ml.

Total: 11 marks

2. (a) Factorize completely:

(i) $4a^2 - 16$

[2]

$$\begin{aligned} &4a^2 - 16 \\ &= 4(a^2 - 4) \\ &= 4(a + 2)(a - 2) \end{aligned}$$

(ii) $3y^2 + 2y - 8$

[2]

$$\begin{aligned} &3y^2 + 2y - 8 \\ &= 3y^2 + 6y - 4y - 8 \\ &= 3y(y + 2) - 4(y + 2) \\ &= (3y - 4)(y + 2) \end{aligned}$$

(b) Solve the simultaneous equations:

$$2x + y = 3$$

$$5x - 2y = 12$$

[5]

$$2x + y = 3 \quad \dots \text{Equation 1}$$

$$5x - 2y = 12 \quad \dots \text{Equation 2}$$

Multiply Equation 1 by 2 to get:

$$4x + 2y = 6 \quad \dots \text{Equation 3}$$

Equation 3 + Equation 2 gives:

$$9x = 18$$

$$x = \frac{18}{9}$$

$$x = 2$$

Substituting $x = 2$ into Equation 1 gives:

$$2(2) + y = 3$$

$$4 + y = 3$$

$$y = 3 - 4$$

$$y = -1$$

$$\therefore x = 2 \text{ and } y = -1$$

- (c) The table below shows corresponding values of the variables x and y , where y varies directly as x .

x	6	10	t
y	3	u	9

Calculate the values of t and u .

[3]

Since y is directly proportional to x , then

$$y \propto x$$

$$y = kx$$

Substituting $x = 6$ and $y = 3$ into $y = kx$ gives:

$$3 = k(6)$$

$$k = \frac{3}{6}$$

$$k = \frac{1}{2}$$

So, we have $y = \frac{1}{2}x$.

When $x = 10$ and $y = u$,

$$u = \frac{1}{2}(10)$$

$$u = 5$$

When $x = t$ and $y = 9$

$$9 = \frac{1}{2}t$$

$$t = 9 \times 2$$

$$t = 18$$

$$\therefore u = 5 \text{ and } t = 18$$

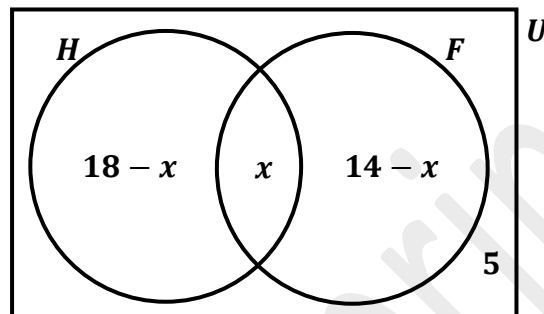
Total: 12 marks

3. (a) The Venn diagram below shows the number of students who study History and French in a class of 30 students.

$U = \{\text{students in the class}\}$

$H = \{\text{students who study History}\}$

$F = \{\text{students who study French}\}$



- (i) Write an expression, in x , in its simplest form, for the TOTAL number of students in the class. [2]

$$\begin{aligned}\text{Total number of students in class} &= 18 - x + x + 14 - x + 5 \\ &= 37 - x\end{aligned}$$

- (ii) State whether the following relationships are true or false: [2]

• $H \cup F = U$ **False**

$$(H \cup F)' = 5 \text{ which is not equal to } 0$$

$$\therefore H \cup F \neq U$$

Hence, the statement given is false.

- $H \cap F' = \emptyset$ False

$$\begin{aligned} H \cap F' &= 18 - 7 \\ &= 11 \end{aligned}$$

So, $H \cap F' \neq \emptyset$.

Hence, the statement given is false.

- (iii) Determine the number of students who study BOTH History and French.

[2]

There are a total of 30 students in the class.

So, we have,

$$37 - x = 30$$

$$x = 37 - 30$$

$$x = 7$$

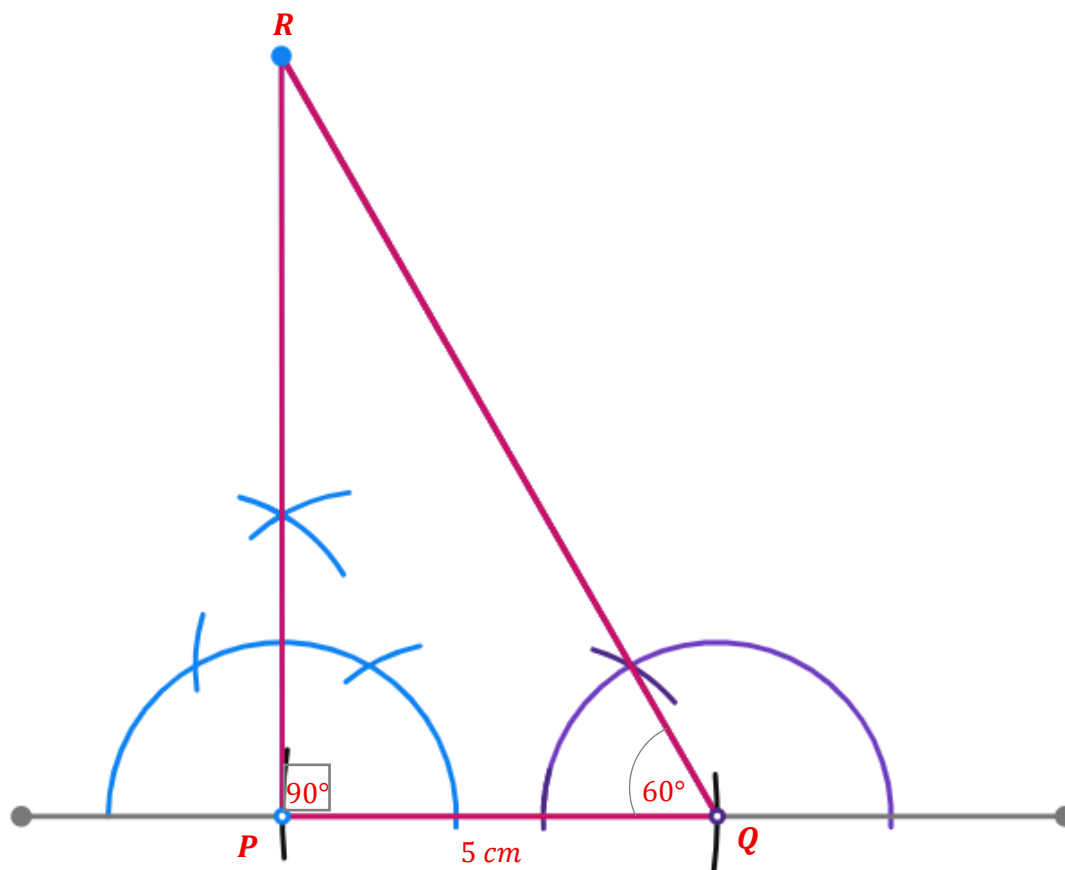
So, $n(H \cap F) = 7$.

$\therefore$ The number of students who study both History and French is 7.

(b)(i) Using a ruler, a pencil and a pair of compasses, construct triangle PQR

with $PQ = 5\text{ cm}$, $\angle PQR = 60^\circ$ and $\angle QPR = 90^\circ$.

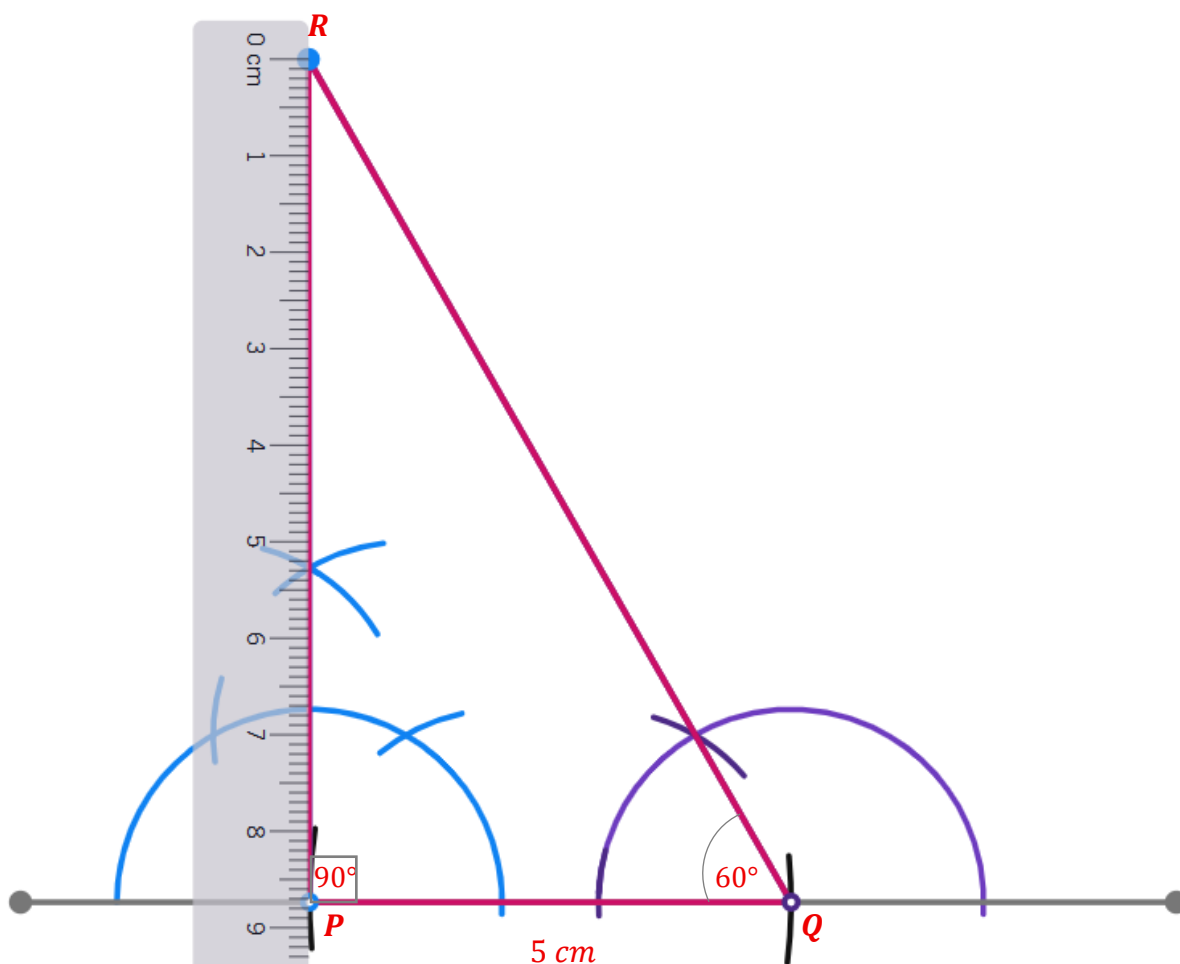
[4]



(ii) Measure and state

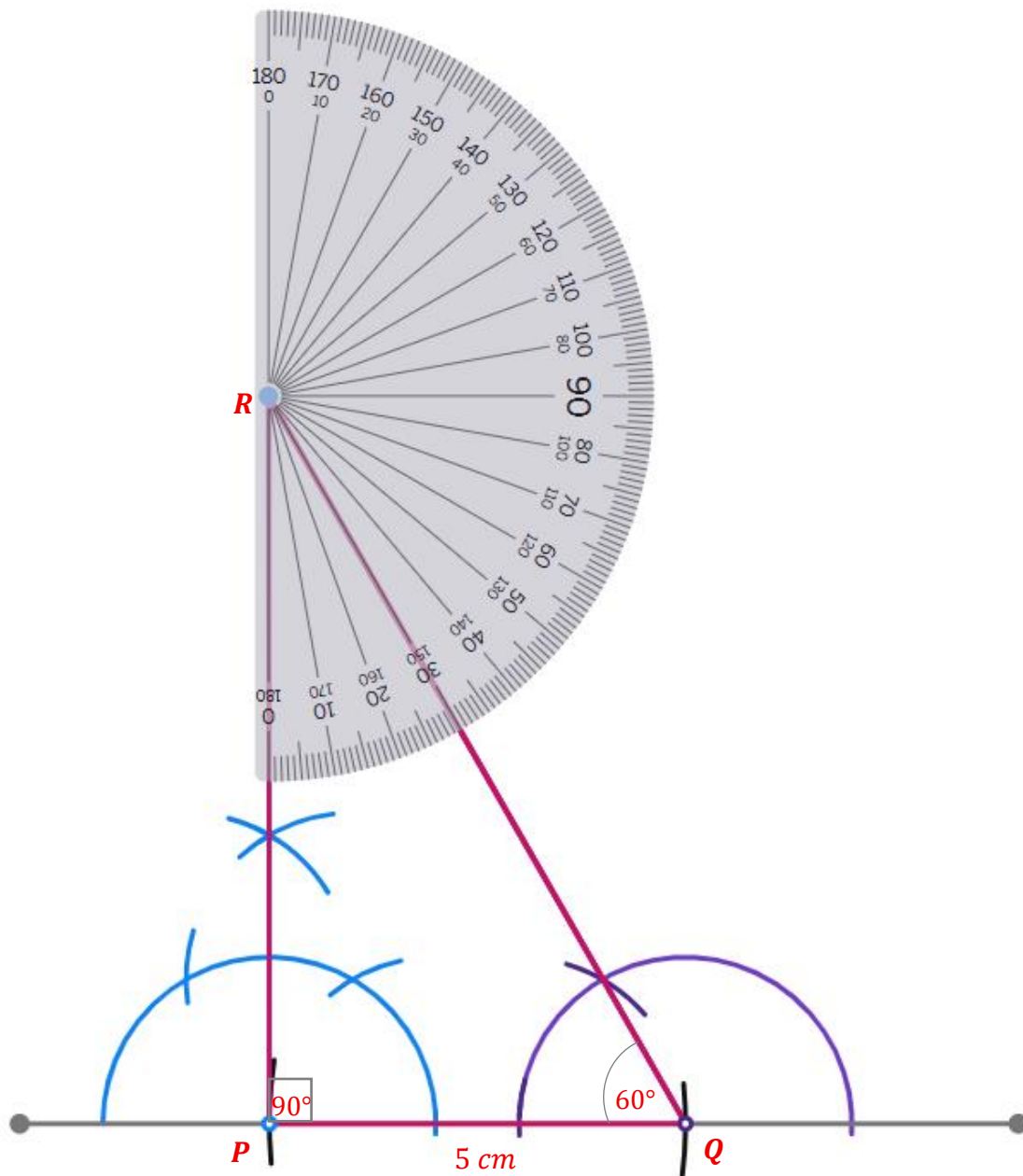
- the length of PR

$PR = 8.75 \text{ cm}$



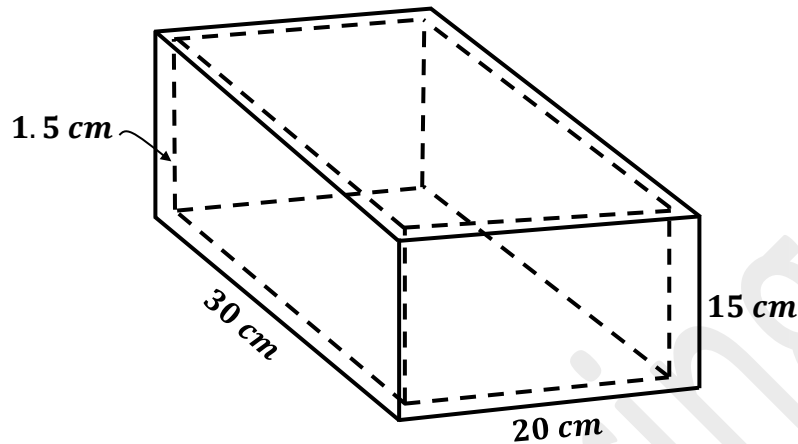
- the measure of $\angle PRQ$

$\angle PRQ = 30^\circ$



Total: 12 marks

4. The diagram below, **not drawn to scale**, shows a silver box with no lid. The sides and bottom of the box are 1.5 cm thick.



- (a) Calculate the volume of the box using the external measurements. [2]

External volume of the box = $l \times w \times h$

$$\text{Volume of the box} = 30 \times 20 \times 15$$

$$\text{Volume of the box} = 9000 \text{ cm}^3$$

$\therefore$ The volume of the box using the external measurements is 9000 cm^3 .

- (b) Complete EACH of the following statements.

- (i) The internal length of the box is $30 \text{ cm} - 2 \times 1.5 \text{ cm} = \dots\dots\dots 27 \text{ cm}$ [1]

$$\text{Internal length} = 30 - 2(1.5)$$

$$= 30 - 3$$

$$= 27 \text{ cm}$$

- (ii) The internal width of the box is 17 cm [2]

$$\text{Internal width} = 20 - 2(1.5)$$

$$= 20 - 3$$

$$= 17 \text{ cm}$$

- (iii) The internal depth of the box is 13.5 cm [1]

$$\text{Internal depth} = 15 - 1(1.5)$$

$$= 15 - 1.5$$

$$= 13.5 \text{ cm}$$

- (c) Calculate the internal volume of the box. [2]

$$\text{Internal volume of the box} = l \times w \times h$$

$$= 27 \times 17 \times 13.5$$

$$= 6196.5 \text{ cm}^3$$

$\therefore$ The internal volume of the box is 6196.5 cm^3 .

- (d) The box is made of silver which has a mass of 10.5 g for each cm^3 . Calculate the mass of the silver box, giving your answer in kg . [3]

$$\begin{aligned}\text{Volume of silver} &= \text{External volume of the box} - \text{Internal volume of the box} \\ &= 9000 - 6196.5 \\ &= 2803.5 \text{ cm}^3\end{aligned}$$

Density of silver is 10.5 gcm^{-3} .

$$\text{Density} = \frac{\text{Mass}}{\text{Volume}}$$

$$10.5 = \frac{\text{Mass}}{2803.5}$$

$$\text{Mass} = 2803.5 \times 10.5$$

$$\text{Mass} = 29436.75 \text{ g}$$

We need to convert this value to kg .

$$1000 \text{ g} = 1 \text{ kg}$$

$$1 \text{ g} = \frac{1}{1000} \text{ kg}$$

$$29436.75 \text{ g} = 29436.75 \times \frac{1}{1000}$$

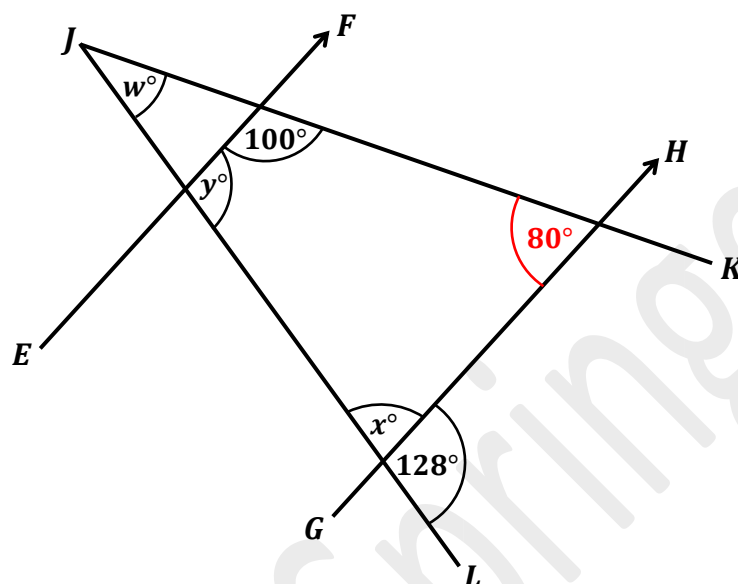
$$= 29.43675$$

$$= 29.4 \text{ kg} \quad (\text{to 3 significant figures})$$

$\therefore$ The mass of the silver box is 29.4 kg .

Total: 11 marks

5. (a) The diagram below, **not drawn to scale**, shows two straight lines, JK and JL , intersecting a pair of parallel lines, EF and GH .



Determine, giving reasons for EACH of your answers, the value of

- (i) x [2]

Angles in a straight line sum to 180° .

$$x = 180^\circ - 128^\circ$$

$$x = 52^\circ$$

- (ii) y [2]

Angle y and 128° are corresponding angles.

Corresponding angles are equal to each other.

$$\therefore y = 128^\circ$$

(iii) w

[2]

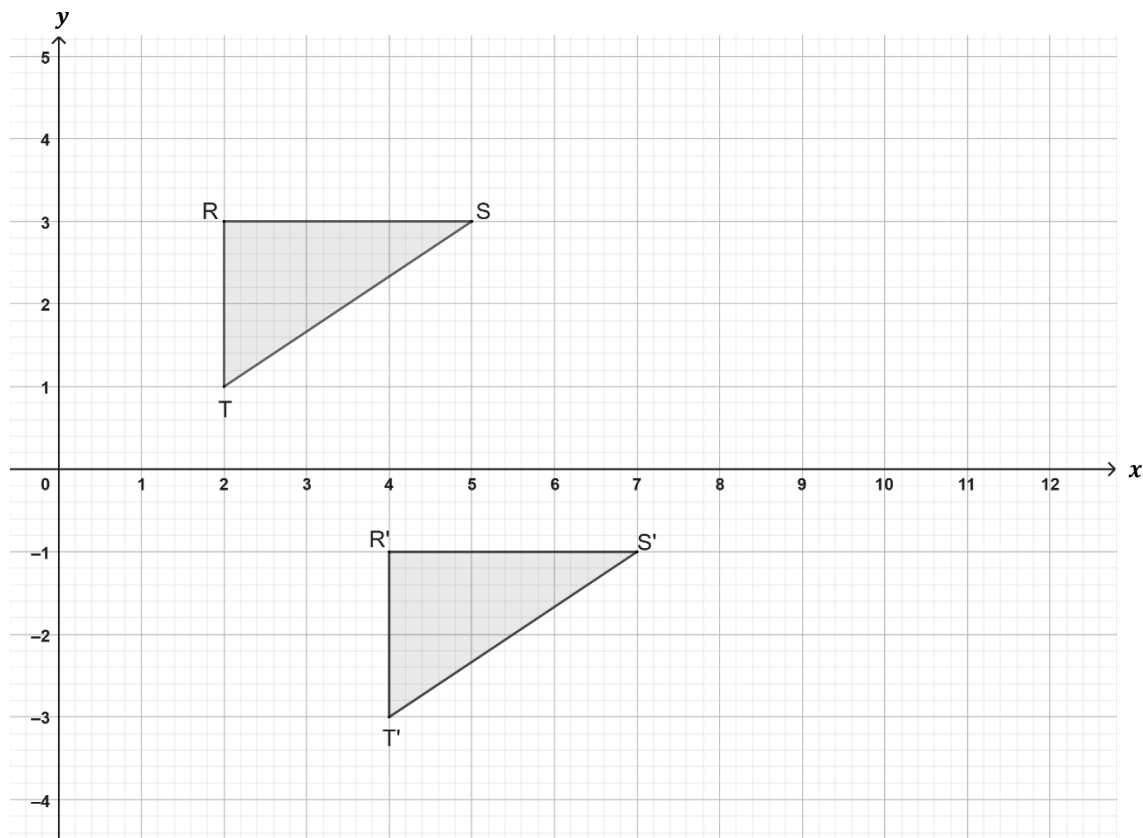
Angles in a quadrilateral add up to 360° .

$$\begin{aligned}\angle JHG &= 360^\circ - (100^\circ + y^\circ + x^\circ) \\ &= 360^\circ - (100^\circ + 128^\circ + 52^\circ) \\ &= 360^\circ - 280^\circ \\ &= 80^\circ\end{aligned}$$

The sum of angles in a triangle is equal to 180° .

$$\begin{aligned}w^\circ + \angle JHG + x^\circ &= 180^\circ \\ w^\circ + 80^\circ + 52^\circ &= 180^\circ \\ w^\circ + 132^\circ &= 180^\circ \\ w^\circ &= 180^\circ - 132^\circ \\ w^\circ &= 48^\circ\end{aligned}$$

- (b) The diagram below shows triangle RST and its image $R'S'T'$ after a transformation.



- (i) Describe FULLY the transformation which maps RST onto $R'S'T'$. [3]

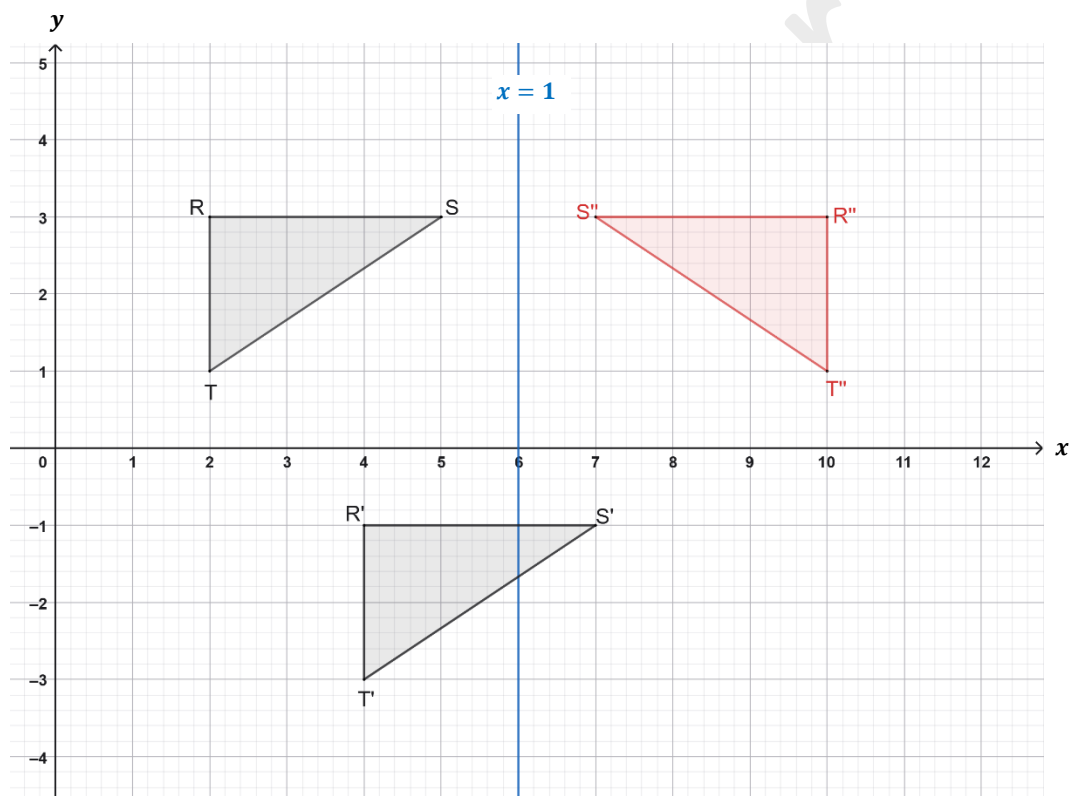
The transformation which maps RST onto $R'S'T'$ is a translation by the vector $\begin{pmatrix} 2 \\ -4 \end{pmatrix}$.

- (ii) Triangle RST is reflected in the line $x = 6$. On the graph above, draw triangle $R''S''T''$, the image of ΔRST , after the reflection.

Write down the coordinates of R'' .

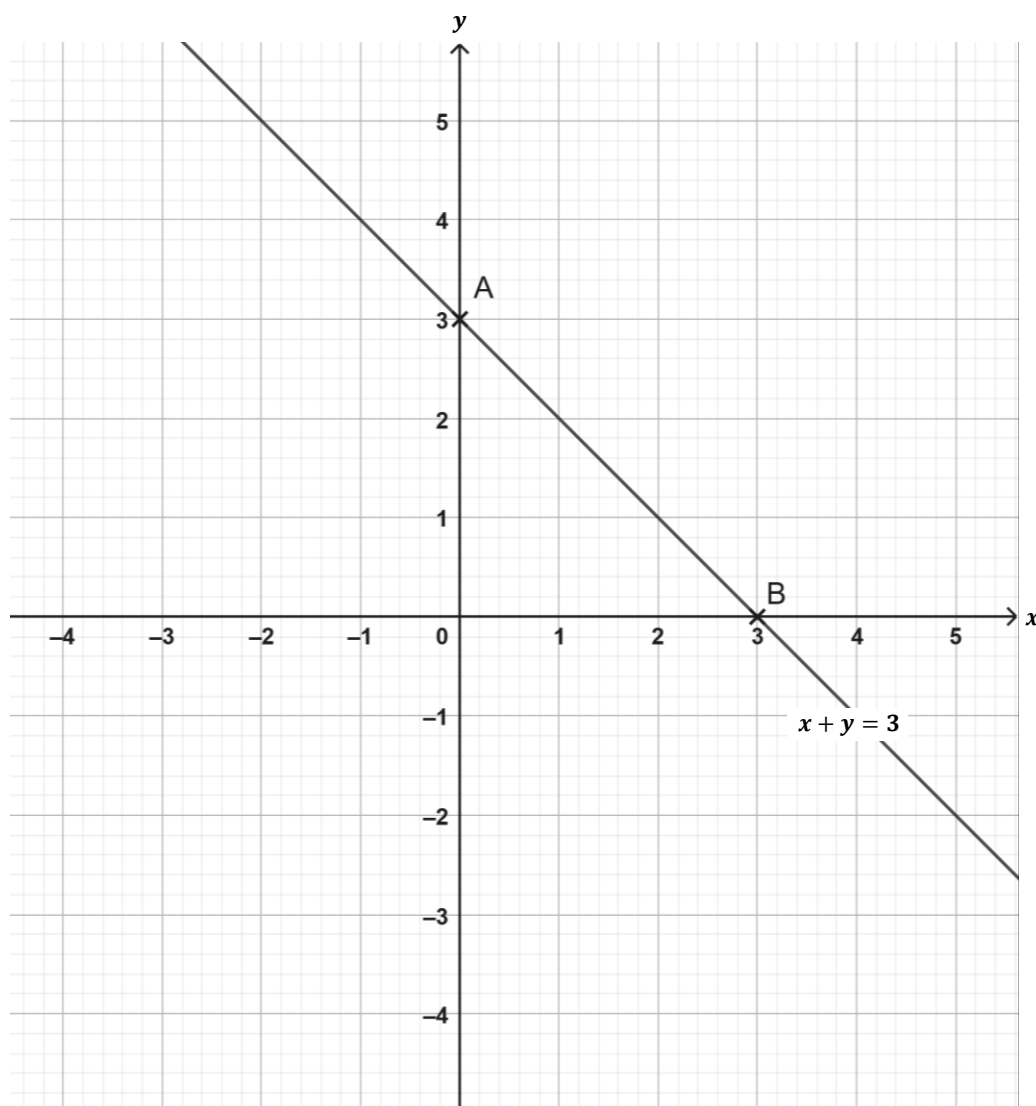
$R'' = (10, 3)$

[3]



Total: 12 marks

6. (a) The diagram below shows the graph of the straight line $x + y = 3$.



Determine the equation of the line which is

- (i) parallel to the line $x + y = 3$ and passes through the origin [2]

Consider the line $x + y = 3$.

Rearranging gives:

$y = -x + 3$ which is in the form $y = mx + c$,

where $m = -1$ and $c = 3$.

Parallel lines have equal gradients.

So, the gradient of the line parallel to the line $x + y = 3$ is $m = -1$.

Now, this parallel line also passes through the origin.

This means that the y -intercept $= 0$.

So, the equation of the parallel line is:

$$y = -x + 0$$

$$y = -x \quad \text{which is in the form } y = mx + c,$$

where $m = -1$ and $c = 0$.

- (ii) perpendicular to the line $x + y = 3$ and passes through the midpoint of AB . [2]

Coordinates at the ends of the line AB : $(0, 3)$ and $(3, 0)$

$$\begin{aligned} \text{Midpoint of } AB &= \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right) \\ &= \left(\frac{0+3}{2}, \frac{3+0}{2} \right) \\ &= \left(\frac{3}{2}, \frac{3}{2} \right) \end{aligned}$$

Gradient of $AB = -1$

The product of the gradients of perpendicular lines $= -1$.

$$\begin{aligned}\text{Gradient, } m &= \frac{-1}{-1} \\ &= 1\end{aligned}$$

Substituting $m = 1$ and point $\left(\frac{3}{2}, \frac{3}{2}\right)$ into the equation of a line:

$$\frac{3}{2} = (1) \left(\frac{3}{2}\right) + c$$

$$c = \frac{3}{2} - \frac{3}{2}$$

$$c = 0$$

$\therefore$ Equation of line is $y = x$.

(b) The function $y = x^2 - 2x - 3$ is defined in the domain $-2 \leq x \leq 4$. The table below shows the corresponding values of y for five selected values of x .

x	-2	-1	0	1	2	3	4
y	5	0	-3		-3		5

(i) Complete the table by calculating and inserting the missing values of y . [2]

$$y = x^2 - 2x - 3$$

When $x = 1$,

$$\begin{aligned} y &= (1)^2 - 2(1) - 3 \\ &= 1 - 2 - 3 \\ &= -4 \end{aligned}$$

When $x = 3$,

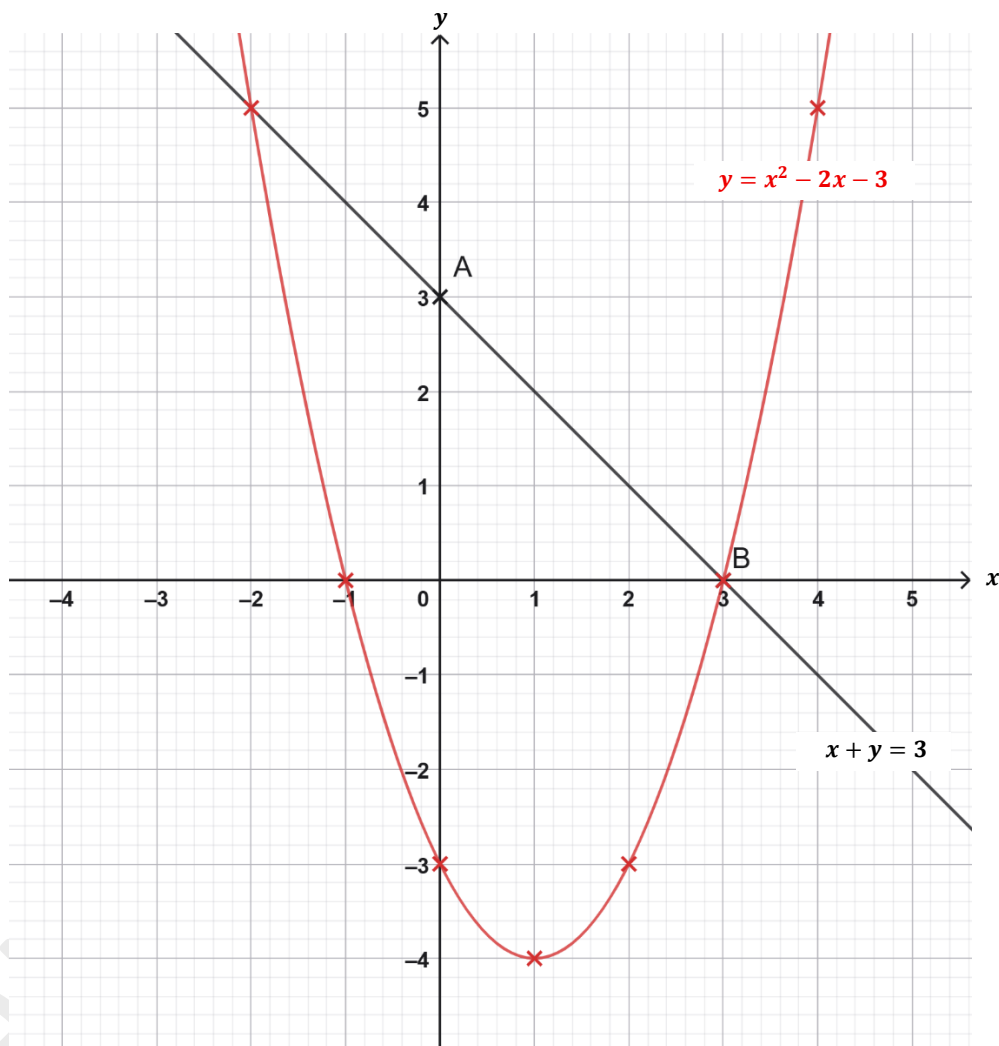
$$\begin{aligned} y &= (3)^2 - 2(3) - 3 \\ &= 9 - 6 - 3 \\ &= 0 \end{aligned}$$

x	-2	-1	0	1	2	3	4
y	5	0	-3	-4	-3	0	5

(ii) On the same axes used in Part (a) on page 16, draw the graph of

$$y = x^2 - 2x - 3.$$

[2]



(iii) Using information from your graph, complete EACH of the following statements.

- The minimum value of $y = x^2 - 2x - 3$ occurs when

$x = \underline{\textcolor{red}{1}} \dots\dots\dots$ [1]

- The values of x for which $x^2 - 2x - 3 = -x + 3$ are

$x = \underline{\textcolor{red}{-2}} \dots\dots\dots$ and $x = \underline{\textcolor{red}{3}} \dots\dots\dots$ [2]

Total: 11 marks

7. Twenty bags of sugar were weighed. The weights, to the nearest kg, are as follows:

3	38	17	33	28
12	43	38	31	30
11	8	23	18	26
50	22	35	39	5

- (a) Complete the frequency table for the data shown above. [4]

Weight (kg)	Tally	Number of Bags
1 – 10	III	3
11 – 20	IIII	4
21 – 30	IIII	5
31 – 40	IIII I	6
41 – 50	II	2

- (b) For the class interval 21 – 30, state:

- (i) the upper class boundary [1]

The upper class boundary of the class interval 21-30 is 30.5.

- (ii) the class width [1]

$$\begin{aligned}
 \text{Class Width} &= \text{Upper Class Boundary} - \text{Lower Class Boundary} \\
 &= 30.5 - 20.5 \\
 &= 10
 \end{aligned}$$

(iii) the class midpoint

$$\begin{aligned}\text{Class Midpoint} &= \frac{UCB+LCB}{2} \\ &= \frac{20.5+30.5}{2} \\ &= 25.5\end{aligned}$$

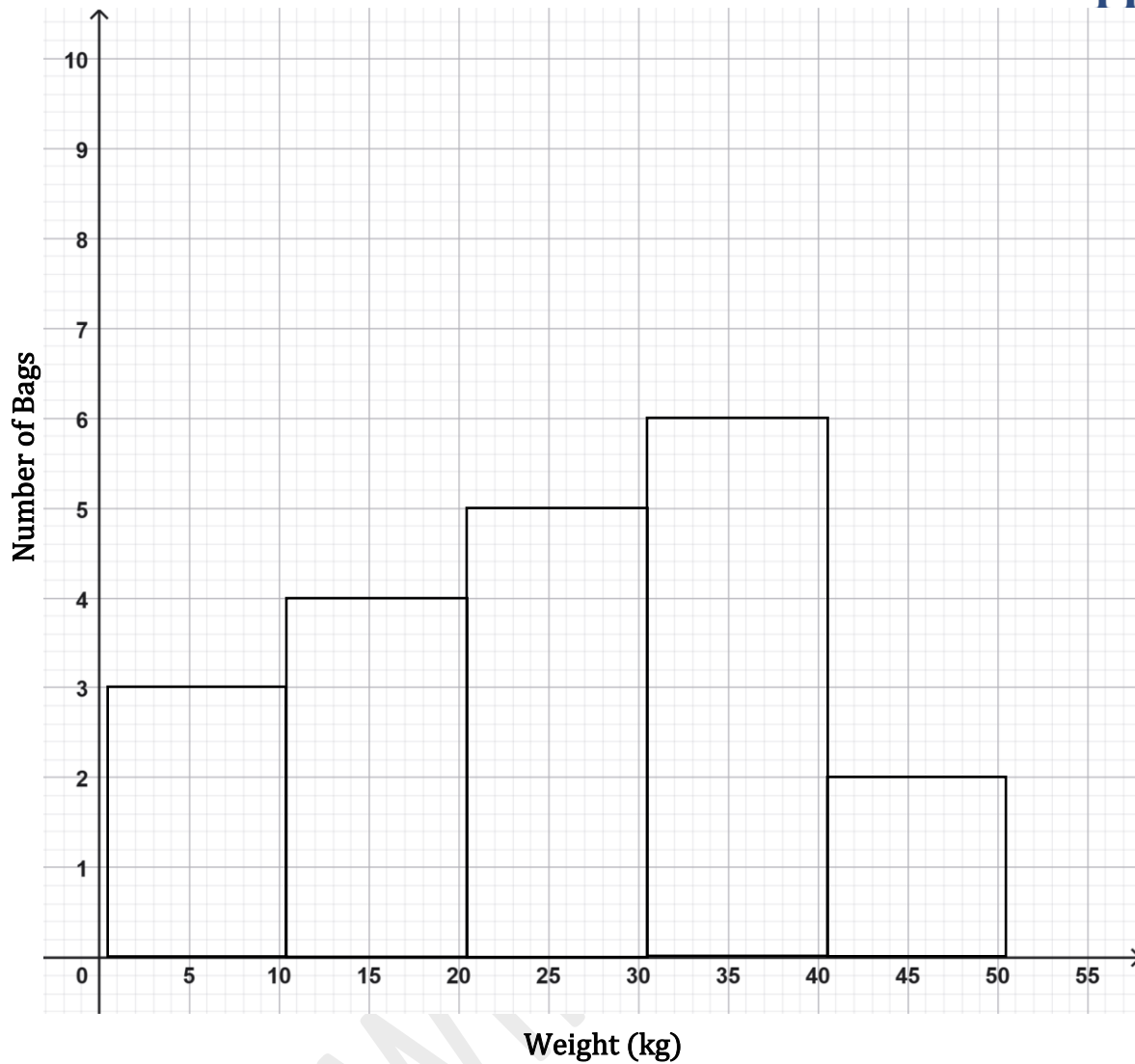
(c) On the grid on page 19, using a scale of **2 *cm* to represent 10 *kg* on the *x*-axis**, and **1 *cm* to represent 1 bag on the *y*-axis**, draw a histogram to represent the data contained in your frequency table above. [4]

Scale Used:

x-axis: 2*cm* represents 10*kg*

y-axis: 1*cm* represents 1 bag

The histogram is shown on the next page.



Total: 11 marks

8. The diagram below show the first three figures in a sequence. Each figure is made up of knots and strings. Each knot connects exactly 3 strings.

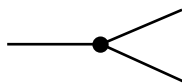


Figure 1

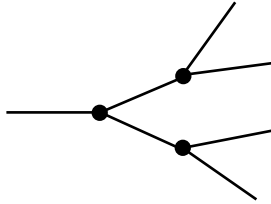


Figure 2

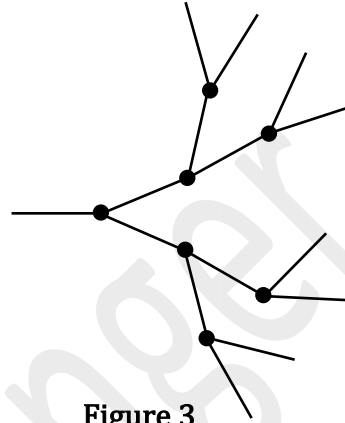


Figure 3

(a) Draw Figure 4 of the sequence.

[2]

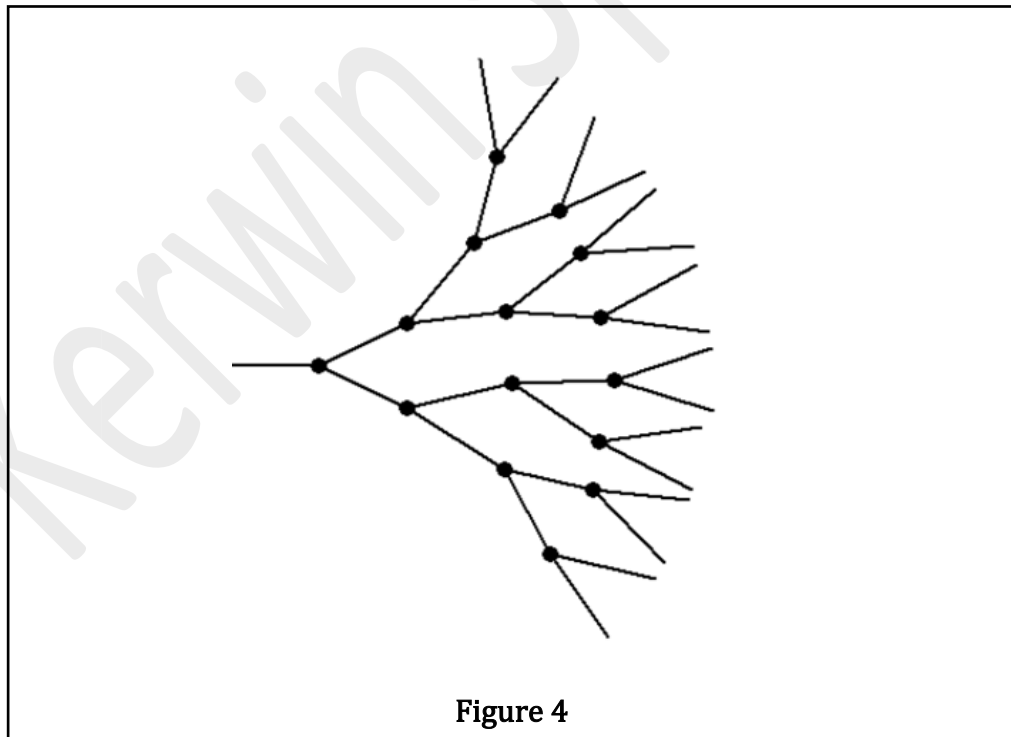


Figure 4

(b) The table below shows the number of knots and strings in each figure used to draw Figures 1, 2 and 3. Complete the table by inserting the missing values in the rows numbered (i), (ii) and (iii).

	Figure Number (N)	Number of Knots (K)	Number of Strings (S)	
	1	1	3	
	2	3	7	
	3	7	15	
(i)	4	<u>15</u>	<u>31</u>	[2]
(ii)	<u>7</u>	<u>127</u>	255	[3]
(iii)	10	<u>1023</u>	<u>2047</u>	[3]

Consider the Figure n .

The number of knots, $K = 2^n - 1$

The number of strings, $S = 2K + 1$

$$\begin{aligned}
 \text{(i)} \quad \text{For Figure 4, number of knots, } K &= 2^4 - 1 \\
 &= 16 - 1 \\
 &= 15
 \end{aligned}$$

For Figure 4, number of strings, $S = 2K + 1$

$$= 2(15) + 1$$

$$= 31$$

(ii) The number of strings, $S = 255$

Since

The number of strings, $S = 2K + 1$

$$255 = 2K + 1$$

$$255 - 1 = 2K$$

$$254 = 2K$$

$$2K = 254$$

$$K = \frac{254}{2}$$

$$K = 127$$

Since

The number of knots, $K = 2^n - 1$

$$127 = 2^n - 1$$

$$127 + 1 = 2^n$$

$$128 = 2^n$$

$$2^7 = 2^n$$

$$\therefore n = 7$$

(iii) For Figure 10, number of knots, $K = 2^{10} - 1$

$$= 1024 - 1$$
$$= 1023$$

For Figure 10, number of strings, $S = 2K + 1$

$$= 2(1023) + 1$$
$$= 2047$$

Total: 10 marks

SECTION II

Answer TWO questions in this section.

ALGEBRA AND RELATIONS, FUNCTIONS AND GRAPHS

9. (a) The functions $f(x)$ and $g(x)$ are defined as $f(x) = 2x - 7$ and $g(x) = x^2 + 1$, respectively.

(i) Write an expression, in terms of x , for EACH of the following:

- $f^{-1}(x)$

$$f(x) = 2x - 7$$

$$\text{Let } y = f(x).$$

$$y = 2x - 7$$

Interchange the variables x and y .

$$x = 2y - 7$$

Make y the subject.

$$2y = x + 7$$

$$y = \frac{x+7}{2}$$

$$\therefore f^{-1}(x) = \frac{x+7}{2}$$

- $g^{-1}(x)$

$$g(x) = x^2 + 1$$

Let $y = g(x)$.

$$y = x^2 + 1$$

Interchange the variables x and y .

$$x = y^2 + 1$$

Make y the subject.

$$y^2 = x - 1$$

$$y = \sqrt{x - 1}$$

$$\therefore g^{-1}(x) = \sqrt{x - 1}$$

- $fg(x)$

$$\begin{aligned} fg(x) &= f[g(x)] \\ &= 2(x^2 + 1) - 7 \\ &= 2x^2 + 2 - 7 \\ &= 2x^2 - 5 \end{aligned}$$

- $(fg)^{-1}(x)$

Let $y = fg(x)$.

$$y = 2x^2 - 5$$

Interchange the variables x and y .

$$x = 2y^2 - 5$$

Make y the subject.

$$2y^2 = x + 5$$

$$y^2 = \frac{x+5}{2}$$

$$y = \sqrt{\frac{x+5}{2}}$$

$$\therefore (fg)^{-1}(x) = \sqrt{\frac{x+5}{2}}$$

(ii) Show that $(fg)^{-1}(5) = g^{-1}f^{-1}(5)$.

$$(fg)^{-1}(x) = \sqrt{\frac{x+5}{2}}$$

$$\begin{aligned}(fg)^{-1}(5) &= \sqrt{\frac{5+5}{2}} \\ &= \sqrt{\frac{10}{2}} \\ &= \sqrt{5}\end{aligned}$$

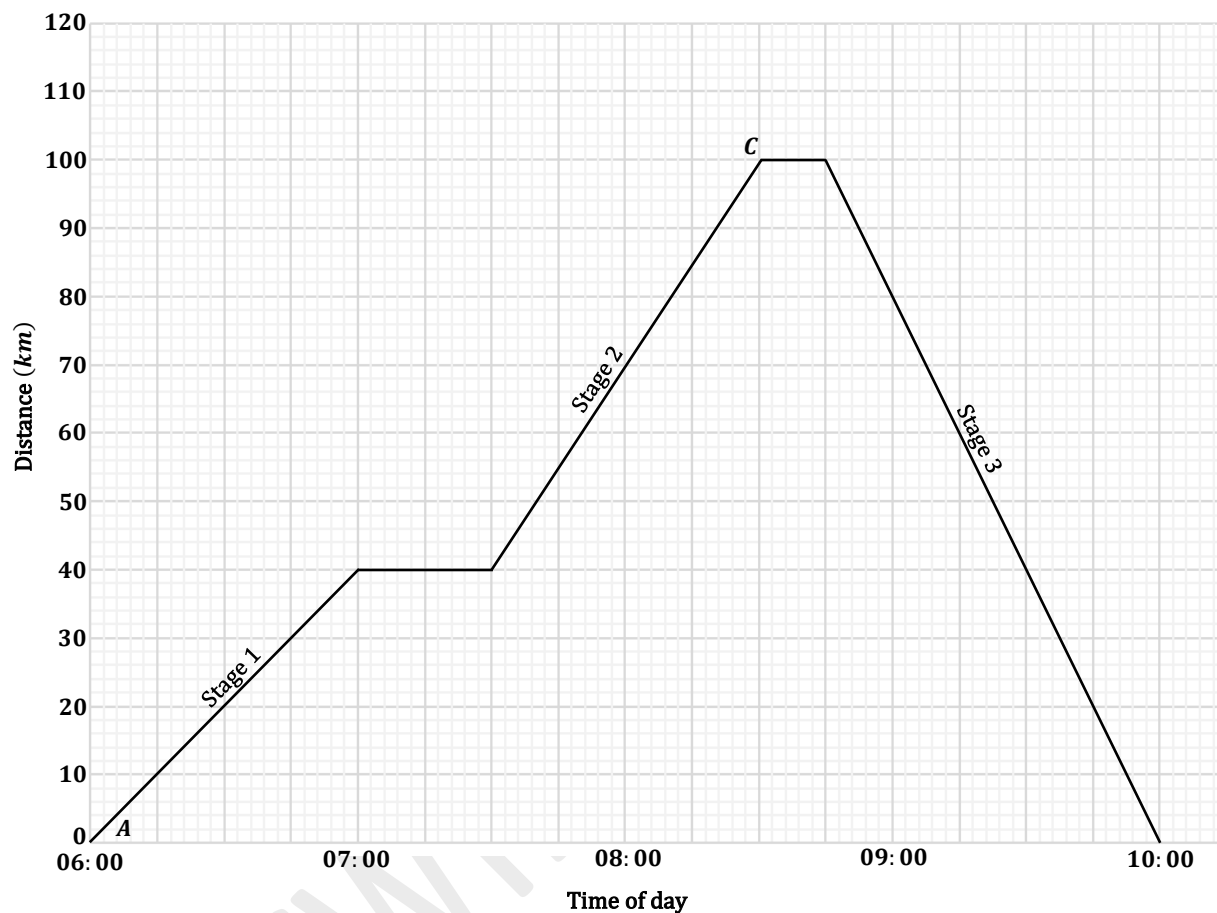
$$\begin{aligned}f^{-1}(5) &= \frac{5+7}{2} \\ &= \frac{12}{2} \\ &= 6\end{aligned}$$

$$\begin{aligned}g^{-1}f^{-1}(5) &= g^{-1}(6) \\ &= \sqrt{6-1} \\ &= \sqrt{5}\end{aligned}$$

$$\therefore (fg)^{-1}(5) = g^{-1}f^{-1}(5)$$

Q.E.D.

- (c) The distance-time graph below shows the three-stage journey of a car travelling from Town A to Town C and back to Town A.



- (i) State the time of day at which the car arrived at Town C.

[1]

The car arrived at Town C at 08:30.

- (ii) Calculate the TOTAL time, in minutes, for which the car stopped during the journey. [2]

The car was at a stop from 07:00 to 07:30 which is 30 minutes and from 08:30 to 08:45 which is 15 minutes.

$$\begin{aligned}\text{The total time the car stopped} &= 30 + 15 \\ &= 45 \text{ minutes}\end{aligned}$$

- (iii) Determine the constant speed of the car during Stage 2 of the journey. [2]

$$\begin{aligned}\text{Constant speed} &= \frac{\text{Distance}}{\text{Time}} \\ &= \frac{100-40}{1 \text{ hr}} \\ &= 60 \text{ kmh}^{-1}\end{aligned}$$

- (iv) Calculate the average speed of the car for the time during which it was moving. [3]

$$\begin{aligned}\text{Total distance} &= 40 + 60 + 100 \\ &= 200 \text{ km}\end{aligned}$$

$$\text{Total time taken} = 1 + 1 + 1\frac{1}{4}$$

$$= 3\frac{1}{4} \text{ hours}$$

$$\text{Average speed} = \frac{\text{Total distance covered}}{\text{Total time taken}}$$

$$= \frac{200}{3\frac{1}{4}}$$

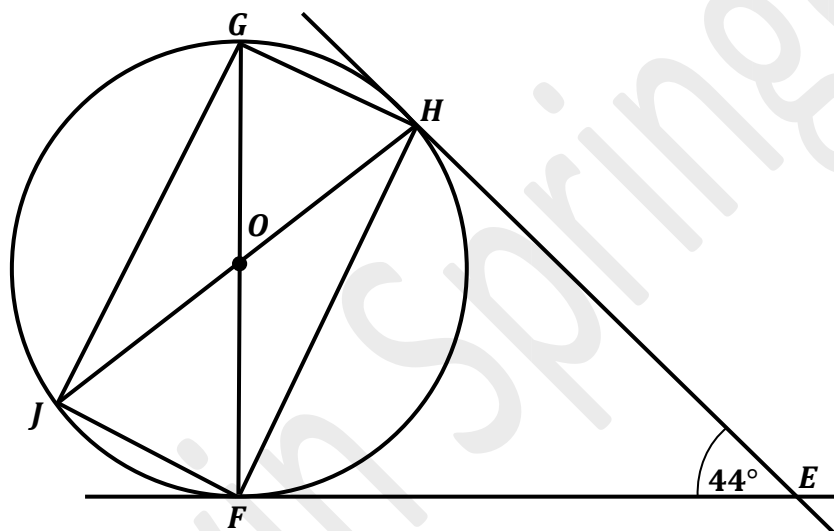
$$= \frac{800}{12}$$

$$= 61.54 \text{ kmh}^{-1}$$

Total: 15 marks

MEASUREMENT, GEOMETRY AND TRIGONOMETRY

10. (a) The diagram below, **not drawn to scale**, shows a circle, centre O . EH and EF are tangent to the circle. FOG and JOH are straight lines. The measure of $\angle FEH = 44^\circ$.



Calculate, **giving reasons for your answer**, the measure of:

- (i) $\angle EHF$

[2]

From point E , two tangents are drawn to the circle at H and F .

Tangents from the same point are equal in length, so:

$$EH = EF$$

This means triangle EHF is an isosceles triangle.

In an isosceles triangle, the base angles are equal, so:

$$\angle EHF = \angle EFH$$

We are given:

$$\angle HEF = 44^\circ$$

The angles in a triangle add up to 180° , so:

$$\angle EHF + \angle EFH = 180^\circ - 44^\circ = 136^\circ$$

Since both angles are equal:

$$\begin{aligned}\angle EHF = \angle EFH &= \frac{136^\circ}{2} \\ &= 68^\circ\end{aligned}$$

(ii) $\angle FGH$

[2]

A radius and a tangent meet at 90° .

So,

$$\angle OHE = 90^\circ$$

$$\angle OFE = 90^\circ$$

In quadrilateral $FOHE$:

$$\begin{aligned}\angle FOH &= 360^\circ - (90^\circ + 90^\circ + 44^\circ) \\ &= 136^\circ\end{aligned}$$

The angle at the circumference is half the angle at the centre.

$$\begin{aligned}\angle FGH &= \frac{136^\circ}{2} \\ &= 68^\circ\end{aligned}$$

(iii) $\angle JHE$ [2]

A radius and tangent meet at 90° .

$$\angle OHE = 90^\circ$$

Since J , O , and H are on the same straight line,

$$\angle JHE = 90^\circ$$

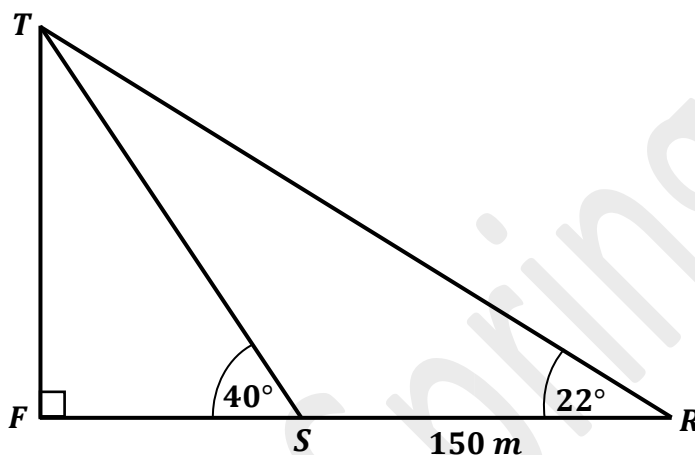
(iv) $\angle JGH$ [2]

JH is a diameter.

The angle in a semicircle is 90° .

$$\angle JGH = 90^\circ$$

- (b) The diagram below, **not drawn to scale**, shows two ships, R and S at anchor on a lake of calm water. FT is a vertical tower. FSR is a straight line and $RS = 150\text{ m}$. The angles of elevation of T , the top of a tower, from R and S , are 22° and 40° respectively. F is the foot of the tower.



Calculate, giving your answer to 1 decimal place where appropriate

- (i) the measure of $\angle RTS$ [1]

Angles in a straight line = 180° .

$$\begin{aligned} \hat{TSR} &= 180^\circ - 40^\circ \\ &= 140^\circ \end{aligned}$$

The sum of the angles in a triangle is equal to 180° .

$$\begin{aligned} \hat{RTS} &= 180^\circ - (140^\circ + 22^\circ) \\ &= 18^\circ \end{aligned}$$

- (ii) the length of ST

Using the sine rule,

$$\frac{ST}{\sin 22^\circ} = \frac{150}{\sin 18^\circ}$$

$$ST = \frac{150 \times \sin 22^\circ}{\sin 18^\circ}$$

$$ST = 181.8 \text{ m (to 1 decimal place)}$$

- (iii) the height of the tower, FT

[3]

$$\frac{TF}{181.83} = \sin 40^\circ$$

$$TF = 181.83 \times \sin 40^\circ$$

$$TF = 116.9 \text{ m (to 1 decimal place)}$$

Total: 15 marks

VECTORS AND MATRICES

11. (a) The position vectors of points A, B and C , relative to the origin O , are

$$\overrightarrow{OA} = \begin{pmatrix} 2 \\ -2 \end{pmatrix}; \overrightarrow{OB} = \begin{pmatrix} 6 \\ 1 \end{pmatrix} \text{ and } \overrightarrow{OC} = \begin{pmatrix} 10 \\ 4 \end{pmatrix} \text{ respectively.}$$

(i) Express in the form $\begin{pmatrix} x \\ y \end{pmatrix}$ the vectors

$$\bullet \overrightarrow{AB} = \begin{pmatrix} 4 \\ 3 \end{pmatrix} \dots\dots\dots$$

$$\bullet \overrightarrow{AC} = \begin{pmatrix} 8 \\ 6 \end{pmatrix} \dots\dots\dots$$

[3]

The position vectors of points A, B and C , relative to the origin O , are $\overrightarrow{OA} =$

$$\begin{pmatrix} 2 \\ -2 \end{pmatrix},$$

$$\overrightarrow{OB} = \begin{pmatrix} 6 \\ 1 \end{pmatrix} \text{ and } \overrightarrow{OC} = \begin{pmatrix} 10 \\ 4 \end{pmatrix} \text{ respectively.}$$

$$\overrightarrow{AB} = \overrightarrow{AO} + \overrightarrow{OB}$$

$$= -\overrightarrow{OA} + \overrightarrow{OB}$$

$$= -\begin{pmatrix} 2 \\ -2 \end{pmatrix} + \begin{pmatrix} 6 \\ 1 \end{pmatrix}$$

$$= \begin{pmatrix} 4 \\ 3 \end{pmatrix}$$

$$\overrightarrow{AB} = \begin{pmatrix} 4 \\ 3 \end{pmatrix} \text{ is of the form } \begin{pmatrix} x \\ y \end{pmatrix}, \text{ where } x = 4 \text{ and } y = 3$$

Now,

$$\begin{aligned}\overrightarrow{AC} &= \overrightarrow{AO} + \overrightarrow{OC} \\ &= -\overrightarrow{OA} + \overrightarrow{OC} \\ &= -\begin{pmatrix} 2 \\ -2 \end{pmatrix} + \begin{pmatrix} 10 \\ 4 \end{pmatrix} \\ &= \begin{pmatrix} 8 \\ 6 \end{pmatrix}\end{aligned}$$

$\overrightarrow{AC} = \begin{pmatrix} 8 \\ 6 \end{pmatrix}$ is of the form $\begin{pmatrix} x \\ y \end{pmatrix}$, where $x = 8$ and $y = 6$

- (ii) Hence, determine whether A , B and C are collinear, **giving the reasons for your answer.** [3]

$$\overrightarrow{AB} = \begin{pmatrix} 4 \\ 3 \end{pmatrix}$$

$$\begin{aligned}\overrightarrow{AC} &= \begin{pmatrix} 8 \\ 6 \end{pmatrix} \\ &= 2 \begin{pmatrix} 4 \\ 3 \end{pmatrix}\end{aligned}$$

$\overrightarrow{AC}$ is a scalar multiple ($=2$) of $\overrightarrow{AB}$.

Hence, $\overrightarrow{AC}$ is parallel to $\overrightarrow{AB}$.

A is a common point. Hence, A and B both lie on $\overrightarrow{AC}$ and A , B and C are collinear.

(b) Determine the value of x for which the matrix $\begin{pmatrix} 3 & x \\ 2 & 4 \end{pmatrix}$ is singular. [2]

$$\text{Let } A = \begin{pmatrix} 3 & x \\ 2 & 4 \end{pmatrix}.$$

$$\begin{aligned} |A| &= (3 \times 4) - (x \times 2) \\ &= 12 - 2x \end{aligned}$$

If A is a singular matrix, then $|A|$ or $\det A = 0$.

$$\therefore 12 - 2x = 0$$

$$2x = 12$$

$$x = \frac{12}{2}$$

$$x = 6$$

(c) N and P are 2×2 matrices such that $N = \begin{pmatrix} 4 & 1 \\ 3 & 2 \end{pmatrix}$ and $P = \begin{pmatrix} 1 & 5 \\ 2 & 1 \end{pmatrix}$.

(i) Determine NP . [1]

$$N = \begin{pmatrix} 4 & 1 \\ 3 & 2 \end{pmatrix} \text{ and } P = \begin{pmatrix} 1 & 5 \\ 2 & 1 \end{pmatrix}$$

$$NP = \begin{pmatrix} 4 & 1 \\ 3 & 2 \end{pmatrix} \begin{pmatrix} 1 & 5 \\ 2 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} (4 \times 1) + (1 \times 2) & (4 \times 5) + (1 \times 1) \\ (3 \times 1) + (2 \times 2) & (3 \times 5) + (2 \times 1) \end{pmatrix}$$

$$= \begin{pmatrix} 4 + 2 & 20 + 1 \\ 3 + 4 & 15 + 2 \end{pmatrix}$$

$$= \begin{pmatrix} 6 & 21 \\ 7 & 17 \end{pmatrix}$$

- (ii) Given that $PN = \begin{pmatrix} 19 & 11 \\ 11 & 4 \end{pmatrix}$, determine whether matrix multiplication is commutative. [1]

We are given that $PN = \begin{pmatrix} 19 & 11 \\ 11 & 4 \end{pmatrix}$.

It was found that $NP = \begin{pmatrix} 6 & 21 \\ 7 & 17 \end{pmatrix}$.

Since $PN \neq NP$, then matrix multiplication is not commutative.

- (iii) Determine N^{-1} , the inverse of N . [2]

$$N = \begin{pmatrix} 4 & 1 \\ 3 & 2 \end{pmatrix}$$

$$\det N = (4 \times 2) - (1 \times 3)$$

$$= 8 - 3$$

$$= 5$$

$$\therefore N^{-1} = \frac{1}{5} \begin{pmatrix} 2 & -(1) \\ -(3) & 4 \end{pmatrix}$$

$$= \frac{1}{5} \begin{pmatrix} 2 & -1 \\ -3 & 4 \end{pmatrix}$$

$$= \begin{pmatrix} \frac{2}{5} & -\frac{1}{5} \\ -\frac{3}{5} & \frac{4}{5} \end{pmatrix}$$

(iv) Hence, calculate the values of x and y for which

$$\begin{pmatrix} 4 & 1 \\ 3 & 2 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}.$$

[3]

$$\begin{pmatrix} 4 & 1 \\ 3 & 2 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$$

$$\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 4 & 1 \\ 3 & 2 \end{pmatrix}^{-1} \begin{pmatrix} 1 \\ 2 \end{pmatrix}$$

$$\begin{pmatrix} x \\ y \end{pmatrix} = N^{-1} \begin{pmatrix} 1 \\ 2 \end{pmatrix}$$

$$\begin{pmatrix} x \\ y \end{pmatrix} = \frac{1}{5} \begin{pmatrix} 2 & -1 \\ -3 & 4 \end{pmatrix} \begin{pmatrix} 1 \\ 2 \end{pmatrix}$$

$$\begin{pmatrix} x \\ y \end{pmatrix} = \frac{1}{5} \begin{pmatrix} (2 \times 1) + (-1 \times 2) \\ (-3 \times 1) + (4 \times 2) \end{pmatrix}$$

$$\begin{pmatrix} x \\ y \end{pmatrix} = \frac{1}{5} \begin{pmatrix} 2 - 2 \\ -3 + 8 \end{pmatrix}$$

$$\begin{pmatrix} x \\ y \end{pmatrix} = \frac{1}{5} \begin{pmatrix} 0 \\ 5 \end{pmatrix}$$

$$\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

$$\therefore x = 0 \text{ and } y = 1$$

Total: 15 marks

END OF TEST

IF YOU FINISH BEFORE TIME IS CALLED, CHECK YOUR WORK ON THIS TEST.